



RITAM MONDAL | 23IM30018

Indian Institute of Technology Kharagpur

Dual Degree (B. Tech and M. Tech) in INDUSTRIAL AND SYSTEMS ENGINEERING



[my github](#)



[Linkedin](#)

portfolio [website](#)

## EDUCATION

| Year | Degree/Exam                                               | Institute                                |
|------|-----------------------------------------------------------|------------------------------------------|
| 2028 | B.Tech and M.Tech (DD), Industrial and System Engineering | Indian Institute of Technology Kharagpur |
| 2022 | Class XII, WBCHSE (87.8%)                                 | Howrah Zilla School, Howrah              |
| 2020 | Class X, WBBSE (91%)                                      | Kona High School, Howrah                 |

## PROJECTS AND EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                          |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| <b>AI-Powered Industrial Knowledge Intelligence Platform   ET AI Hackathon 2026 (Round 2 Qualifier)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Jun'26 - Present</b>  |
| <ul style="list-style-type: none"><li>Architecting an AI platform for unified industrial knowledge discovery across documents, maintenance records and SOPs</li><li>Designing a multi-agent system with Copilot, RCA, Compliance, and Lessons Learned agents</li><li>Developing a RAG pipeline using vector databases for context-aware industrial query answering</li><li>Building document ingestion workflows with OCR, entity extraction, and semantic indexing</li><li>Implementing compliance intelligence to identify regulatory gaps and support audit readiness</li><li>Engineering maintenance intelligence for RCA support and predictive maintenance recommendations</li></ul>                                                                                  |                          |
| <b>Optimizing Delivery ETAs with Graph-Based Network Intelligence   Summer Project   IIT Guwahati   <a href="#">GitHub</a></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Apr'26 – Jun'26</b>   |
| <ul style="list-style-type: none"><li>Engineered a directed weighted transportation graph with <b>1,657 nodes</b> and <b>2,783 edges</b> using <b>144K+ Delivery trip records</b>.</li><li>Extracted graph analytics features including betweenness centrality, degree metrics, corridor delays, and node2vec embeddings</li><li>Developed Graph-Enhanced XGBoost models, improving <b>R<sup>2</sup> by 167% (0.15 → 0.40)</b> and <b>reducing MAE by 10.7% (16.26 → 14.53 min)</b></li><li>Formulated corridor intelligence framework for FTL vs Carting route analysis and operational decision support</li><li>Deployed end-to-end Streamlit <a href="#">application</a> integrating OSRM API, Folium visualization, ETA prediction, and delay-risk assessment</li></ul> |                          |
| <b>MLOps Custom Lung X-Ray model for Pneumonia Prediction   Self Project (ongoing)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Jun'26 – Present</b>  |
| <ul style="list-style-type: none"><li>Architected RobustXrayNet, a <b>custom deep learning network in PyTorch</b> featuring Global Average Pooling (GAP), and a dropout-regularized classifier head</li><li>Mitigated severe class imbalance using stratified dataset splitting, label smoothing, and class-weighted CrossEntropyLoss to eliminate prediction bias, stabilizing overall classification <b>accuracy at 84% with an F1-score of 0.89</b></li><li>Building model serving workflows using BentoML for versioned deployment</li><li>Designing an automated ingestion and storage pipeline integrating AWS S3 for secure dataset and artifact handling and Docker based infrastructure</li></ul>                                                                  |                          |
| <b>Web Based Attention Monitoring through eye tracking (IoT project)   Academic Project (IM39204)   Mamata Jenamani, IIT Kharagpur   <a href="#">github</a></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>Jan'26 – Apr'26</b>   |
| <ul style="list-style-type: none"><li>Integrated WebGazer.js frontend with backend APIs, enabling real-time gaze data collection and dashboard visualization.</li><li>Developed gaze tracking pipeline using MediaPipe, generating time-series attention signals from webcam streams</li><li>Designed visual analytics (<b>heatmaps, scanned paths, velocity plots</b>) to interpret user attention patterns and behavior.</li><li>Deployed backend and enabled data export (CSV/JSON) with HTTPS-ready setup for live webcam-based applications.</li><li>Containerized the application using <b>Docker</b>, enabling consistent environment setup, dependency management, and easier deployment across systems.</li></ul>                                                  |                          |
| <b>Optimizing Revenue Leakage &amp; Profitability in the Hospitality Sector   Summer Project   IIT Guwahati   <a href="#">Github</a></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>May'25 – Jun'25</b>   |
| <ul style="list-style-type: none"><li>Architected a full-stack hospitality analytics solution on 32K+ booking records to identify revenue leakage drivers across multiple booking channels</li><li>Performed data preprocessing and EDA using Python, Pandas, sea born and Google Colab workflows</li><li>Designed interactive Power BI dashboard tracking <b>15+ operational KPIs</b> including ADR, RevPAR, Occupancy %, and Realization %</li><li>Simulated “late booking penalty” scenario — <b>reduced revenue leakage by 49%</b></li><li>Recommended pricing and channel optimization strategies: weekday bundling, platform shift, non-refundable policy</li></ul>                                                                                                   |                          |
| <b>Portfolio Optimization in stock Market   Academic Project - IM29204   prof Avishek Sharma   IIT Kharagpur   <a href="#">gihub</a></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Feb'25 - April'25</b> |
| <ul style="list-style-type: none"><li>Optimized portfolio allocation across 10 equities using Sharpe Ratio maximization under investment constraints</li><li>Processed 2 years of historical stock data and evaluated risk-return tradeoffs using MATLAB and Python</li><li>Applied constrained optimization techniques to maximize risk-adjusted returns</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                         |                          |
| <b>Forest Cover Type Prediction (Green Vision ML)   Online learning Course Project  PW Skills   <a href="#">github</a></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <b>June'25 – July'25</b> |
| <i>Machine Learning   FastAPI   AWS S3   Docker   Full-Stack Deployment</i> <ul style="list-style-type: none"><li>Predicted predominant forest cover type from cartographic features using USFS Region 2 dataset</li><li>Built ML pipeline with feature engineering, model training &amp; evaluation (RandomForest, XGBoost)</li><li>Designed full-stack architecture with database integration and FastAPI web interface</li><li>Connected storage with AWS S3 bucket for dataset and model persistence</li><li>Containerized with Docker and deployed web app for interactive prediction</li></ul>                                                                                                                                                                        |                          |
| <b>Rock Paper Scissors GUI Game with Audio-Visual Effects   Intern Project – Vault of Code   <a href="#">(github)</a></b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Jun'25</b>            |
| <ul style="list-style-type: none"><li>Developed an interactive GUI-based game with audio-visual feedback using Tkinter and Pygame</li><li>Implemented game state management, score tracking, and multi-round gameplay logic</li><li>Designed responsive user interactions with custom graphics and sound effects</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                          |

## SKILLS AND EXPERTISE

- Programming:** Python, C++, C, R
- Machine Learning:** Regression, Classification, Ensemble Learning, XGBoost, Random Forest, SVM, ARIMA, SARIMA
- Deep Learning & Computer Vision:** CNNs | Transfer Learning | TensorFlow | PyTorch | YOLO | MediaPipe | Object Detection | Object Tracking | Image Segmentation
- Generative AI:** Retrieval-Augmented Generation (RAG) | LangChain | Ollama | Prompt Engineering | Transformer
- Data Analytics & Visualization:** Exploratory Data Analysis (EDA) | Statistical Learning | Time Series Analysis | Excel Solver | Matplotlib | Seaborn
- Deployment & MLOps:** FastAPI | Streamlit | Docker | AWS S3
- Version Control:** Git | GitHub

## COURSEWORK INFORMATION

Probability and Statistics | Statistical Learning with Applications | Production Planning and Control | Forecasting | Simulation | Operations Research I & II | Quality Design and Control | Supply Chain Management | Management of Inventory Systems |

TRAINING

**PW Skills - Data Science and Generative AI**

- Completed training in Python, Machine Learning, Deep Learning, NLP , RAG and Generative AI
- Learned feature engineering, model development, deployment, and MLOps fundamentals
- Gained hands-on experience with TensorFlow, PyTorch, FastAPI, Docker, and AWS

**Python Development Internship | Vault of Code | Jun'25**

- Completed internship focused on Python application development
- Developed interactive desktop applications using Tkinter, PIL, and Pygame
- Applied software development principles including event-driven programming and user interface design

CERTIFICATES

- C/C++ (PW skills) , Machine Learning (AlmaBetter),
- Python,Algorithmic Problem solving intermediate(Hacker Rank)
- Quant quest,Kshitij-IIT kgp'25 algorithmic trading certificate (by Unstop)

POSITIONS OF RESPONSIBILITY AND EXPERIENCE

**Kshitij, Techno management fest of IIT kharagpur | Kshitij Campus Affiliate**

Oct'23 - August'24

- Coordinated publicity and outreach activities for Kshitij 2024, contributing to engagement across 60,000+ participants
- Publicised the fest in the state of Jharkhand (Dhanbad, Ranchi, Deoghar) ,Bihar (patna), West Bengal (Kolkata, Hooghly) . Also make editors to publish the fest in their respective newspaper of Gano Shakti, Telegraph, Ananda bazar, Times of India

**Student Mentor | Physics Wallah**

April'24 - August'24

Mentored and guided JEE around 350 students for their preperation

AWARDS AND ACHIEVEMENTS

- Solved **400+ Data Structures and Algorithms** problems across **LeetCode** and HackerRank
- Achieved LeetCode **Contest Rating of 1800+**
- Earned 5★ HackerRank Badge in Problem Solving
- Qualified for Round 2 **of ET AI Hackathon 2026** and currently developing an AI-powered Industrial Knowledge Intelligence platform
- Secured All India Rank of **5435** in **JEE Advanced 2023**(out of 1,90,000 candidates) and All India Rank of 20819 in JEE Mains 2023(out of 13,00,000 candidates).
- WBCHSE (class XII) - 87.8%
- WBBSE (class X) - 91%